

Quality Power Electrical Equipments Limited (QPOWER)

India's Emerging HVDC & FACTS Component Supplier to the Global Grid Transition

Sector	Power Transmission & Distribution – HVDC / FACTS Components
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Note: This report is compiled entirely from the research inputs, notes and figures supplied by the client. No external data has been added. Where source notes contained differing figures for the same metric, both figures are presented and flagged accordingly.

1. Executive Summary

Quality Power Electrical Equipments Limited (“Quality Power”) is a specialised power engineering company operating in the transmission segment of the electricity value chain (Generation → Transmission → Storage → Distribution). The company is a B2B supplier of high-voltage grid components and subsystems — principally reactors, capacitors and filters — used inside High Voltage Direct Current (HVDC) converter stations and Flexible AC Transmission System (FACTS) installations. It is not a system integrator; it supplies critical, custom-engineered sub-components to global OEMs such as Hitachi Energy, ABB and GE Vernova, which in turn build converter stations for large grid projects.

The investment case rests on three converging trends: (i) India's renewable energy build-out, which requires long-distance transmission infrastructure where HVDC has a structural advantage; (ii) a global HVDC capex cycle across India, the United States and Europe; and (iii) a “China+1” shift in OEM sourcing that favours pre-qualified, non-Chinese suppliers such as Quality Power.

Quality Power holds one of only a small number of OEM-certified positions in India for reactors, capacitor banks and filters, alongside BHEL and (in one data point) TBEA — a Chinese-linked manufacturer. Revenue mix is shifting toward the higher-margin Power Products (HVDC) segment, which has grown from 34% to 52% of revenue over three years, while the company simultaneously executes an acquisition-led strategy (Sukrut Electric, Mehru, Endoks, Nebeskie Labs) to build a multi-product, single-vendor proposition for OEM customers. The company is currently operating at 100% capacity utilisation and has a consolidated order book of approximately Rs. 1,300 crore, equivalent to roughly 12 months of revenue visibility.

Key risks include commodity price volatility (copper and silicon steel), the possibility of OEMs vertically integrating into sub-component manufacturing, low-cost Chinese competition on exports, working-capital intensity, customer concentration, and dependence on continued government renewable-energy policy support.

2. Industry Backdrop: Why HVDC Matters

India currently transmits electricity using two technologies. High Voltage Alternating Current (HVAC) handles 90–95% of transmission today; it has a lower initial cost but higher power losses over distance. High Voltage Direct Current (HVDC) handles only 5–10% of transmission currently; it carries a higher upfront cost but delivers materially lower power losses and a stronger long-term return on investment. HVDC is not intended to replace HVAC — the efficient design mix uses HVDC for long-distance bulk transmission and HVAC for shorter, local distribution, analogous to highways carrying bulk cargo while local roads handle last-mile delivery.

2.1 The Economics of HVDC vs. HVAC

For a representative 1,000 km transmission line, source estimates put HVDC capex at approximately Rs. 6,500 crore versus Rs. 4,300 crore for HVAC — a premium of roughly Rs. 2,200 crore. However, HVDC saves an estimated Rs. 1,300–2,100 crore per year in reduced transmission losses, implying a payback period on the capex premium of just 1–2 years. Over a 30-year asset life, cumulative loss savings are estimated at Rs. 39,000–63,000 crore on a single corridor.

- HVDC can carry more MW on the same corridor width than HVAC (with a conversion loss of roughly 2% each way, AC to DC and back).
- HVDC does not require frequency matching, allowing it to connect asynchronous grids seamlessly.
- HVDC offers better control and stability, with less breakage and lower maintenance over time.

Power Grid Corporation has announced two additional HVDC projects for FY27. Newer large projects are also scaling in voltage/capacity requirements — from roughly 1 GW toward 7–8 GW in some cases (Reliance cited as an example) — a scale at which building a small number of HVDC lines becomes preferable to building many HVAC lines with attendant synchronisation complexity.

3. Company Overview and Value Chain Positioning

Quality Power sits within the Transmission stage of the power value chain (Generation → Transmission → Storage → Distribution). Its core focus is on high-voltage grid components and subsystems used in HVDC and FACTS networks, and on power quality and grid-stability solutions. The company operates as a B2B supplier to global OEMs and EPC players rather than as an end-system integrator.

Management notes that Quality Power is one of only a small number of OEM-certified component manufacturers in India for this category, alongside BHEL and TBEA (a China-linked manufacturer).

3.1 Order Flow – How Revenue Reaches Quality Power

Orders originate with government / Power Grid project announcements, flow to EPC contractors, and then to OEM system integrators, who in turn procure sub-components:

- Government / Power Grid announces an HVDC project.
- EPC Contractor (e.g., L&T, Adani, Sterlite) is awarded the project.
- OEM / System Integrator (Hitachi Energy, GE Vernova, ABB) assembles the converter station.
- Quality Power supplies smoothing reactors, AC/DC filters and capacitor banks into that converter station.

A parallel path exists where Power Grid awards fully in-house manufacturing to BHEL; however, source notes describe BHEL as having well-documented issues with damages, delays, quality and labour inefficiency, supported by audit data. Hitachi Energy, by contrast, outsources its components, and Quality Power sits within that outsourced supply chain. Within the converter station itself, the converter transformer is typically supplied by BHEL / TARIL / TBEA, converter valves by the OEM's own division (e.g., Hitachi, Siemens), and control & protection systems by specialists such as NR Electric or Siemens automation — while smoothing reactors, AC/DC filters and capacitor banks are supplied by Quality Power.

Note: Quality Power confirms it is itself an OEM customer for certain inputs — it outsources semiconductor chips, cooling systems, and IGBT/thyristor valves used in its FACTS and HVDC products; this is not something the underlying OEMs state explicitly.

3.2 Key OEM Customers

Customer	Market Capitalisation	Order Book
Hitachi Energy	Rs. 1.4 lakh crore	Rs. 30,000 crore
ABB India	Rs. 1.5 lakh crore	Rs. 10,500 crore
GE Vernova T&D India	Rs. 1.15 lakh crore	Rs. 13,100 crore

Source: Client research notes. These OEMs carry substantial order books of their own, implying a sustained downstream pipeline of component demand for Quality Power.

4. Product Portfolio

Quality Power's products are mostly custom-engineered, high-specification parts rather than commodity products, split across two technology categories.

4.1 HVDC Components (DC Transmission)

- Smoothing Reactors – control power flow and protect the system.
- AC/DC Filters – clean power quality.
- Capacitor Banks – support voltage and efficiency.

4.2 FACTS Components (AC Transmission Enhancement)

FACTS is a smart control layer added onto existing AC transmission infrastructure. If traditional AC transmission is a highway, FACTS is the traffic-signal system, lane management and speed regulation installed on that same highway to carry more “traffic” without building new roads.

- STATCOMs – a more advanced version of SVCs, offering faster response and better stability, at a higher cost.
- SVCs (Static VAR Compensators) – enhance grid stability.
- Series Capacitors – increase power transmission capacity by reducing reactance.

Using the highway analogy for HVDC specifically: reactors act as speed breakers and stabilisers, capacitors act as boosters that keep power flowing efficiently, and filters act as the pollution-control system. All are direct inputs into HVDC converter stations, supplied to OEMs such as Hitachi Energy and ABB, and are directly linked to HVDC and renewable grid capex cycles.

4.3 Products in Development / Adjacent Categories

- HVDC-grade CTC Magnetic Wire – a premium raw material input used in reactors for both HVDC and FACTS applications, and required for high-voltage transformers; previously outsourced, now being brought in-house, which management expects to improve profitability.
- Surge Capacitors – protect high-voltage transformers from voltage spikes, lightning strikes and switching surges; necessary for GIS technology and currently a bundled product in development.
- Battery Energy Storage Systems (BESS) context – BESS functions as a power bank supplying electricity during peak demand and is complementary to FACTS and HVDC systems, since transmission projects require some buffer storage (currently only a few hours' worth). Quality Power's related offering covers EMS (control panels, monitoring, energy optimisation, software/communication systems) and power conversion (inverter panels, power electronics, control and switching systems).

On transformer manufacturing specifically, management notes the company is not currently entering full power-transformer manufacturing; it already produces instrument transformers through Mehru and is targeting subsystems of power transformers, consistent with its inorganic growth pathway.

5. Business Segments and Revenue Mix

Quality Power reports two revenue segments, and the mix has shifted meaningfully toward the higher-margin category over the past three years:

Segment	3 Years Ago	Most Recent
Power Products (custom-engineered HVDC components)	34%	52%
Power Quality Equipment (FACTS / standard components)	64%	46%

Interim year mix: Power Products 41%, Power Quality Equipment 56% (three-year progression: 34%–41%–52% and 64%–56%–46% respectively).

Management frames this shift as evidence that the company is consciously transitioning from a volume-driven electrical equipment player toward a high-growth, high-margin HVDC/FACTS component supplier.

On execution speed, project timelines for large OEMs such as Hitachi Energy typically span 2–3 years. Quality Power executes FACTS projects in approximately 12–13 months, with HVDC project timelines broadly comparable to OEM timelines at 2–3 years.

6. Growth Strategy: Acquisitions and Backward Integration

Quality Power has pursued a deliberate acquisition strategy to position itself at multiple points of the value chain and to shortcut the lengthy OEM qualification process, which typically takes 5–7 years.

6.1 Sukrut Electric (50% stake acquired for Rs. 5.24 crore)

Sukrut was previously owned by a German parent company that had already completed the OEM approval process — a rigorous, multi-year qualification exercise. Sukrut's confirmed customer list evidences that those approvals remain intact, giving Quality Power pre-validated access to OEM supply chains. Sukrut sits in the middle of the transformer value chain: Raw Material (magnet wire) → Components (Sukrut) → Finished Equipment (Mehru's instrument transformers). One note in the research indicates management expects Sukrut's revenue to grow 7–8x over 3–4 years, supported by existing ties to multiple global OEMs.

6.2 Mehru (51% stake acquired for Rs. 120 crore)

Mehru's core products are bushings and instrument transformers. It has a partnership with Hyosung T&D India (a Korean conglomerate with approximately \$3.9 billion in revenue) for Gas-Insulated Switchgear (GIS) insulator/instrument-transformer technology. GIS is reliable, space-efficient (roughly 5x more space-efficient than conventional switchgear) and complex to manufacture, creating high entry barriers that protect margin; it is the preferred technology for metro rail, data centres, offshore wind and urban grids. Mehru is expected to be among the first Indian companies to manufacture GIS transformers for global markets, with commercialisation expected within roughly 12 months. The global GIS market is estimated to grow at a 7–8% CAGR through 2030. Mehru turned EBITDA-positive for the first time during the period covered by these notes, and expansion is underway. The remaining 49% of Mehru continues to be held by the Sharma family, the original promoters.

6.3 Endoks (51% stake, Turkey-based)

Endoks gives Quality Power a direct export presence in Europe. Export mix as a share of consolidated revenue was 60% in FY24, moderating to 40% in FY25. Endoks Enerji benefits from a 0% effective tax rate, driven by R&D-linked incentives and exemptions, which management says enhances profitability and cash flow at the consolidated level.

6.4 Nebeskie Labs (26% stake, acquired for Rs. 1.67 crore at a 6.4x revenue multiple)

Nebeskie Labs is focused on AI analytics, IoT monitoring and edge computing, embedding digital intelligence into Quality Power's smart-grid solutions for real-time monitoring, predictive maintenance and enhanced grid performance. Its revenue has grown at an estimated ~100% CAGR; Deshpande Startups is noted as a pre-seed investor in the company.

6.5 The Single-Vendor Proposition

OEM customers prefer sourcing multiple products from a single vendor, since this reduces procurement hassle, cuts the number of separate qualification processes required, and consolidates accountability. The combination of Sukrut, Mehru, Endoks, in-house CTC magnetic wire and the in-development surge capacitor gives Quality Power an increasingly multi-product profile across the transformer and reactor value chain, spanning both import and export markets.

7. Capacity and Expansion

Quality Power is currently operating at 100% capacity utilisation, which management frames as both evidence of demand strength and the reason expansion is now underway.

- Sangli Global Coil Factory – expansion subject to environmental clearance.
- Global Engineering and Technology Centre, Sangli – a new R&D facility.
- Mehru expansion – already EBITDA-positive for the first time.
- Cochin expansion – fully complete and operational.
- Instrument transformer manufacturing facility in Turkey – under consideration / early-stage idea, not yet committed.

On capacity scale, one research note puts Sangli's total capacity at approximately Rs. 1,500 crore of revenue as of Q3 FY26, with a roughly 9x expansion pathway implying an eventual capacity of approximately Rs. 13,500 crore. Mehru's capacity is estimated at approximately Rs. 300 crore for FY26, rising to approximately Rs. 450–500 crore with new capex.

8. Financial Snapshot

Consolidated revenue is reported to have grown roughly 3x, from approximately Rs. 337 crore to approximately Rs. 900–1,000 crore, with profit growing approximately 4x over the same period. The consolidated order book stands at approximately Rs. 1,300 crore, providing roughly 12 months of revenue visibility.

8.1 Entity-Level Revenue (approximate, per research notes)

Entity	Approximate Revenue
Quality Power (standalone)	Rs. 290–300 crore
Sukrut	Rs. 4–5 crore
Mehru	Rs. 400 crore (FY25: Rs. 258 cr; FY26E: ~Rs. 300 cr)
Endoks	Rs. 150–160 crore

Mehru quarterly revenue (FY26): Q1 Rs. 61 cr, Q2 Rs. 79 cr, Q3 Rs. 83 cr, FY estimate ~Rs. 300 cr. Endoks quarterly revenue (FY26): Q1 Rs. 91 cr, Q2 Rs. 74 cr, Q3 Rs. 149 cr. Entity-level figures as noted are approximate and, when summed, are broadly consistent with the ~Rs. 900–1,000 crore consolidated revenue figure cited elsewhere.

8.2 EBITDA Margins by Business Line

Business Line	EBITDA Margin
Coil product factory	34%
Endoks	~30%
Mehru	16.4%
Consolidated (current)	~20%
Consolidated (forward guidance)	~25%, on better product mix and expansion

8.3 Exports

Standalone exports are reported at approximately Rs. 67 crore, with consolidated exports (including Endoks' European operations) at approximately Rs. 225 crore. Management notes the export story remains in its early

stages, and that Quality Power's integrated India–Europe presence enables delivery across Europe typically within hours to a maximum of 3–4 days.

9. Market Opportunity (TAM)

Note: the underlying research notes contain more than one estimate for India's addressable market and Quality Power's current market share; both are reproduced below rather than reconciled, since the report is restricted to client-supplied data.

9.1 India

- HVDC component TAM: one estimate puts current TAM at Rs. 1,500 crore, growing at an 18% CAGR to Rs. 3,500 crore; a second estimate puts current TAM at Rs. 4,500 crore, growing at the same 18% CAGR to Rs. 14,300 crore. A further note references a narrower Rs. 1,100–1,500 crore TAM range.
- FACTS component TAM: approximately Rs. 3,000 crore.
- Quality Power's current India market share: estimates range from approximately 12–13% to 15–20%, and in one note as high as 20–25%, depending on the segment and year referenced (FY26).
- Across the various estimates, the consistent conclusion in the research notes is that Quality Power could roughly double its revenue within two years even without any additional market-share gain, purely on underlying market growth.

9.2 Global

- Global TAM: USD 15–20 billion, described as largely untapped by Quality Power today.
- Global FACTS market: USD 1.72 billion in 2024, projected to grow at a 7.8% CAGR through 2030.

10. Demand Drivers by Geography

10.1 India

- 500 GW renewable energy target by 2030.
- Green Energy Corridor Phase 2.
- Interstate transmission expansion.
- Offshore wind policy rollout.
- Regulatory mandate requiring industrial and commercial sectors to source 35% of power from renewables, rising to 40% within 2–3 years — a direct boost to HVDC infrastructure demand.
- India's total installed generation capacity is approximately 475–520 GW (figures vary slightly across notes), of which roughly 50% belongs to the private sector; private players prioritise reliability, efficiency and profitability, which tends to tilt contracts toward faster, higher-quality suppliers such as Quality Power.
- Generation mix referenced in notes: non-renewable (coal) ~70%; renewable – hydro ~10%, solar ~8%.
- The major unsolved problem in Indian power is storage of renewable generation. As this is addressed, renewables could rise to as much as 50% of total generation from approximately 15–20% today — a scaling path that directly benefits green-corridor and HVDC transmission infrastructure.

Renewable energy in India is mostly developed in remote, isolated locations, which is precisely the use case where HVDC's long-distance efficiency advantage is strongest — the two trends are mutually reinforcing. Power Grid has announced two more HVDC projects in its near-term pipeline.

10.2 Named / Referenced Project Pipeline

- Bhadla – Fatehpur: ~Rs. 25,000 crore (Hitachi + Adani + BHEL)
- Khavda – Nagpur: ~Rs. 10,000 crore (Hitachi + BHEL)
- Ladakh HVDC Project: ~Rs. 20,700 crore
- Mumbai HVDC Infeed
- KPS III HVDC Project
- Raigarh → Pugalur → Thrissur
- Mundra → Mohindergarh
- Talcher → Kolar
- Andaman HVDC link (subsea)
- India–Sri Lanka HVDC (proposed)
- Rihad – Dadri (referenced without further detail in source notes)

10.3 United States

- Only about 3% of current US transmission is HVDC.
- USD 70 billion has been specifically allocated for grid modernisation.
- The AI data-centre boom is creating new energy demand that the existing grid cannot absorb (illustrative data point: a standard Google search consumes about 0.3 Wh of power versus roughly 2.7 Wh for an AI-based search).
- US customers are described as prioritising quality and timely execution over tariff/price considerations, meaning Quality Power's products are not treated as commodity items in that market, with minimal to no direct competition noted.

10.4 Europe

- €584 billion committed to grid expansion by 2030.
- Multiple explicit HVDC projects already announced.
- In 2024, renewable power projects totalling 1,700 GW across 16 European countries were stuck in grid-connection queues — more than six times Germany's total installed generation capacity; HVDC is described as the fastest available solution to this bottleneck.
- The broader industry is running at higher capacity utilisation globally, consistent with a high-demand environment.
- European pricing is reported to run approximately 30–35% higher than equivalent Indian pricing.

11. Competitive Landscape

11.1 Global Competitors

- Trench Group (Germany) – order book of approximately Rs. 14,000 crore.
- Coil Innovation (Austria) – already has an India presence.
- Chinese manufacturers – estimated 30–35% global market share; viewed primarily as an export-market threat.

11.2 Indian Competitors – HVDC

- BHEL – full HVDC system manufacturer, but with documented execution issues (damages, delays, quality, labour, inefficiency).
- TARIL – transformers and reactors.
- TBEA – transformers and reactors; has just entered capacitors and filters.

11.3 Indian Competitors – FACTS

- CG Power – STATCOM/SVC and reactors.
- BHEL – full HVDC system manufacturer.

11.4 The China Question

Raw materials — copper and silicon steel — together make up roughly 50% of total product cost. China holds an advantage on commodity pricing, while India holds an advantage on labour cost; on a pure cost basis, China wins. However, two structural forces are noted as pushing back against Chinese share gains:

- China+1 strategy: OEMs are actively diversifying supply chains away from China, with European OEMs in particular described as having lost trust in single-source Chinese supply.
- OEM qualification lock-in: becoming an approved supplier to an OEM such as Hitachi or ABB takes a minimum of 5–7 years. Quality Power's acquired entities came pre-approved, whereas a new Chinese entrant would need to restart that qualification clock.

Overall, customers are described as prioritising quality, reliability and speed of execution over headline cost.

12. Key Risks

- Commodity price volatility – copper and silicon steel are major input costs.
- OEM self-entry into sub-component manufacturing – the biggest structural risk if a large customer such as Hitachi or ABB decides to bring production in-house.
- Low-cost Chinese competition, particularly in export markets.
- Working-capital constraints – a 12-month order-book visibility is favourable, but execution requires ongoing capital.
- Customer concentration – heavy dependence on a small set of large OEM customers.
- Policy-reliant demand – renewable-energy targets and HVDC capex are government-driven, so policy shifts matter.
- Environmental clearance risk – the Sangli factory capacity expansion is pending environmental clearance.

13. Investment Thesis

Quality Power sits at the intersection of three converging trends: India's renewable-energy push, the global HVDC infrastructure build-out, and the structural shift away from single-source Chinese supply chains. The company manufactures the unglamorous but essential components — reactors, filters and capacitor banks — that go inside the converter stations built by OEMs such as Hitachi Energy and ABB.

In India, Quality Power is positioned as the dominant niche sub-component player in this category at present, with no directly comparable domestic competitor of similar scale and focus. Its acquisition strategy — Sukrut, Mehru, Endoks and Nebeskie Labs — has given it pre-approved access to OEM supply chains that would otherwise take 5–7 years to qualify for organically, while simultaneously building a multi-product, single-vendor value proposition across the transformer and reactor value chain.

The company is running at 100% capacity utilisation while expansion projects are underway at Sangli, Cochin and Mehru. Revenue mix continues to shift toward the higher-margin HVDC (Power Products) segment, and consolidated EBITDA margin is guided from approximately 20% toward approximately 25% as product mix and scale improve. The order book provides roughly 12 months of revenue visibility, key OEM customers carry order books in the tens of thousands of crores, and the addressable market is estimated to be growing at an 18% CAGR in India alone, with a largely untapped USD 15–20 billion global opportunity.

The principal risks to monitor are commodity-cost volatility, the possibility of OEM vertical integration into sub-components, and Chinese competition in export markets — but the OEM-approval moat, first-mover positioning in GIS technology, and the multi-geography (India–Europe–US) demand backdrop represent genuine structural advantages in a market that, per the research notes reviewed, is only in the early stages of scaling.

14. Disclaimer

This note has been compiled solely from research notes, figures and commentary supplied by the client. It has not been independently verified against company filings, exchange disclosures, or other primary sources, and no figures beyond those supplied have been added. Where the source notes contained internally inconsistent figures for the same metric (e.g., TAM size, market share), both figures have been retained and flagged rather than reconciled. This document is for internal research purposes only and does not constitute investment advice or a recommendation to buy or sell any security.